REFERENCE MODEL · WP-02

Extending the SAP semantic layer inside the client's boundary

A reference-architecture description of how the SAP semantic layer can be extended with a client-owned reasoning surface, without replicating financial data out of SAP and without a second definition of any measure. Not a licensed-product replacement.

Executive summary

This paper describes a reference architecture, validated at a mid-market SAP customer, for extending the SAP semantic layer with an in-house AI reasoning surface delivered from infrastructure the client already owns. The reference build proved one point: a semantic access point into SAP DataSphere is sufficient to base an in-house AI reasoning surface on SAP financial data, with SAP's governance preserved and without a second copy of the underlying transaction data.

The pattern is transferable to any SAP customer whose different groups want or need conversational access to SAP data and whose CIO wants SAP's semantic-layer governance preserved. It is not a replacement for Analysis for Office, SAP Analytics Cloud, or Databricks-shaped replication paths — all valid SAP migration or extension patterns. It sits alongside as the pattern for organisations that want their AI reasoning capability inside their own tenant, on their own compute, with the SAP semantic layer as the governed access boundary.

What the reference build validates

A conversational reporting workspace inside the client's Microsoft Teams tenant. A business user asks for a report in plain language or picks one from a catalogue; the platform runs an approved query live against SAP DataSphere and returns an interactive, exportable report in seconds.

A governed AI: standard reports run versioned, pre-approved SQL; the model may only edit named, bounded blocks of that SQL; every run is validated, traced, and stored as an immutable version.

Three exports from the same data: a deterministic Excel workbook, an AI-composed executive Word summary, and an executive slide deck in the structure the board already uses.

An open-source core with a modest metered cost base — order-of-magnitude USD 150–400 per month depending on observability choices — and no per-seat BI licences.

The rule that everything else follows

SAP DataSphere is the semantic layer and the only source of financial truth. Nothing in the platform copies, caches, or re-implements financial data or business definitions. A report is a live, controlled query; the result is held in a temporary in-memory workspace for the life of one request, then written out as a versioned snapshot file. The small application database only records which snapshot is current. This is what makes the numbers reconcilable, the AI auditable, and the platform portable.

1. Reference architecture

1.1 · The four moving parts

The reference build has four parts, and only one of them is customised per client:

The semantic access point. SAP DataSphere, exposing the client's chosen financial views through its OData catalogue. This is the governed access boundary; everything else reads through it.

The reasoning agent. A LangGraph-based router/executor/synthesiser that translates business questions into bounded queries against the semantic layer. The agent may only edit named, pre-approved query blocks; it cannot author arbitrary SQL. Reused unchanged across engagements.

The ephemeral analysis layer. A DuckDB layer that materialises the query result for the life of one request. Nothing persists beyond the session. The versioned snapshot file is what gets retained.

The delivery surface. A Microsoft Teams tab, registered as an Entra app inside the client's tenant. The user experience is a chat window and a report catalogue. Reused unchanged across engagements.

Only the semantic access point is per-client work. The other three are the same code on every engagement. The engineering effort is short because the shape is fixed.

1.2 · What is not in the architecture

There is no standing copy of SAP transaction data. There is no second definition of any measure. There is no vendor-hosted LLM endpoint carrying SAP data across the client boundary. There is no per-seat BI licence. There is no data pipeline that runs on a schedule and produces a warehouse the client then has to govern separately.

2. Delivery clock

The reference retrofit closed in five calendar weeks: about three weeks of client security review (serial, before the build could start), then two weeks of engineering. At the end of week five the platform ran the same recurring finance reports in Teams that the incumbent BI surface produced, at parity against a closed prior period.

The point is not that AI accelerated the build. The point is that once security clears, three of the four parts of the engine are reused unchanged from the reference architecture and only the semantic access point needs authoring for the specific client's DataSphere shape.

3. Portability

The reference build runs today in an Azure tenant. The application code (React, FastAPI, LangGraph, DuckDB) is cloud-agnostic and every managed service has an AWS or on-prem equivalent. Any SAP customer can adopt the same pattern in their preferred cloud without rebuilding the SAP-side definitions and without a second replication of financial data.

4. Where this pattern fits

This reference model is not a competitor to any SAP-licensed product. It is the pattern for organisations that want:

- AI reasoning capability inside their own tenant, on their own compute.
- SAP's semantic-layer governance preserved as the access boundary.
- A short engineering clock and a modest metered cost base.
- The freedom to change cloud or hosting arrangement later without rebuilding.

Where any of those constraints do not apply, licensed BI or replication-based patterns are more appropriate. This paper does not argue against them.

5. Adoption is a separate question

The reference build delivered the technical capability in weeks. Whether a client then retires an incumbent BI surface is a separate adoption decision, not a technical one. At the reference client the decision was deferred; the pattern still holds. This paper describes the pattern; the client-side adoption work is a distinct engagement.

6. About Steel Cloud Solutions

Steel Cloud Solutions is a boutique enterprise-software consultancy. Our SAP practice is client-side advisory — training, integration evaluation, project planning, and specialist technical staffing — for enterprise IT teams running or evaluating SAP. We are not an SAP channel partner, and we do not sell or resell the reference-model stack described in this paper. We help clients evaluate whether the pattern fits their landscape, plan the implementation, and staff the specialist work — via their existing SI or directly, whichever the client prefers.

To scope an evaluation of this pattern for your landscape, email **contact@steelcloudsolutions.com**.